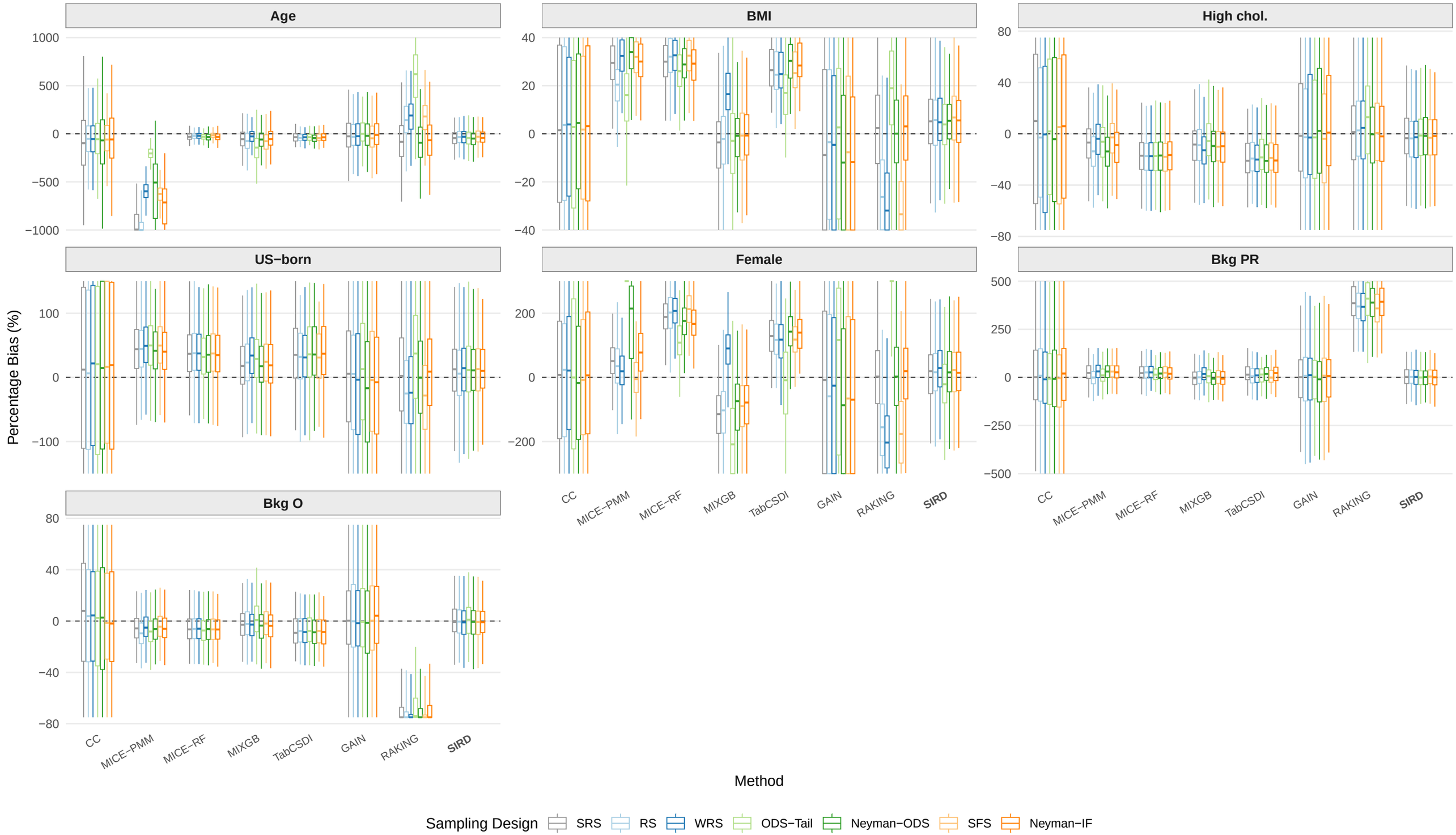Nutrition simulation : Phase−One Covariates (Binomial)